

Homework 4: Neural networks and PINNs

1. Feedforward Networks + AD + Backprop

Use a simple feedforward neural network to approximate the solution to the 1D Poisson equation

$$-u''(x) = f(x), \quad x \in (0, 1),$$

with Dirichlet boundary conditions $u(0) = 0, u(1) = 0$.

- Let $f(x) = \pi^2 \sin(\pi x)$, so the exact solution is $u(x) = \sin(\pi x)$.
- Train a small network (e.g., 2–3 hidden layers, tanh activation).
- Implement the loss using automatic differentiation to compute $u''(x)$.
- Do NOT use existing packages (e.g., PyTorch, Jax, or others) for training, rather
 - Manually code the forward pass.
 - Manually compute gradients using your own implementation of backpropagation (matrix calculus).
 - Update weights using mini-batch stochastic gradient descent

Provide the following *deliverables*

- Hand-written derivation of backpropagation for one of the network layers.
- Python or Julia code implementing the network and training loop.
- Plot comparing the NN solution and exact solution.
- Discuss the following questions:
 - Why does AD give you derivatives for free?
 - How is AD different from backprop?

Rationale: The problem forces you to implement a minimal neural network solver from scratch, so that you *feel* what AD and backprop are doing under the hood, rather than using the existing packages as a black box.

2. Physics-Informed Neural Networks (PINNs)

Use a PINN to solve the time-dependent heat equation

$$u_t = u_{xx}, \quad x \in (0, 1), \quad t \in (0, 1),$$

with initial and boundary conditions

$$u(x, 0) = \sin(\pi x), \quad u(0, t) = u(1, t) = 0.$$

- Construct a single network $u_\theta(x, t)$.
- Train using a PINN loss:
 - PDE residual (interior points)
 - Boundary loss
 - Initial condition loss
- Use automatic differentiation to compute both u_t and u_{xx} .
- Compare to analytical solution:

$$u(x, t) = e^{-\pi^2 t} \sin(\pi x).$$

Provide the following *deliverables*

- Plot PDE residual over training.
- Plots of the predicted solution vs analytical solution at various times.
- Where does the PINN struggle (e.g., stiffness, boundary layer, collocation density)? why?

Rationale: This problem showcases the core ideas of PINNs: Embedding physics in the loss function, balancing residual terms, and understanding why time-dependent PDEs require careful sampling.